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Intelligent Image Recognition Based Automatic Modeling System for Power Equipment Equivalent Circuits

Qiwen Ye, Yu Chen, Zong Deng, and Zhengzheng Liu*

***Abstract* -- Equivalent circuit modeling of power equipment is a crucial foundation for power system analysis, fault diagnosis, and monitoring equipment health. Traditional circuit modeling methods rely on manual identification and calculations. These approaches are inefficient and prone to errors when handling complex circuits. This paper proposes an automatic modeling and simulation system for equivalent circuits of power equipment based on intelligent image recognition. The system uses deep learning technology for automatic component recognition in circuit diagrams and applies image processing algorithms to detect node connection relationships. It automatically establishes Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) equation systems—standard electrical circuit principles to describe current and voltage flows—and then solves circuit frequency-domain characteristics using state-space equations. The system features a triple verification framework. It cross-validates the results from intelligent recognition and automatic calculations with simulation results from manually constructed Multisim models and automatically generated Simulink simulation models via a JSON (JavaScript Object Notation) interface. Using the equivalent circuit of a transformer winding as an example, experimental results show that the system can accurately identify repetitive unit structures, extract circuit features, and establish mathematical models. The high-frequency impedance spectra calculated by the intelligent method agree with results from manual simulation in Multisim and automatic modeling in Simulink. This validates the intelligent modeling method's effectiveness and accuracy. The system automates circuit modeling, providing an efficient and intelligent solution for a wide range of power equipment.**

***Index Terms*-- Circuit image recognition, Equivalent circuit modeling, State space analysis**

I. INTRODUCTION

As the scale of power systems continues to expand and their complexity increases, the precise modeling of power equipment has become one of the key technologies for ensuring the safe and stable operation of power grids. Accurate equivalent circuit models not only serve as the foundation for power system analysis and calculation but also provide a crucial basis for equipment fault diagnosis, health status assessment, and predictive maintenance. Especially in the modeling of critical power equipment such as transformers, reactors, and transmission lines, the accurate extraction of equivalent circuit parameters directly impacts the simulation accuracy and operational safety of power systems [1-2]. Among these, frequency response analysis (FRA) is an important method for diagnosing the mechanical integrity of transformers [3-4]. Its working principle is as follows: when a broadband excitation signal (typically above 1 kHz) is applied to the transformer, the transformer windings exhibit complex electrical characteristics that can be equivalent to a distributed parameter circuit network composed of resistor R, inductor L, and capacitor C [5-6]. The parameters of these circuit components are closely related to the physical state of the windings. When mechanical deformation or insulation aging occurs in the transformer, the corresponding circuit parameters change, thereby altering the system's frequency response characteristics [7]. By comparing the differences between the frequency response curves of the normal state and the current state, fault detection and localization can be achieved [8]. The state-space method provides a solid mathematical theoretical foundation for frequency response analysis (FRA) [9]. By establishing a state-space model describing the dynamic characteristics of the transformer, the relationship between frequency-domain responses and changes in physical parameters can be analyzed, providing a scientific basis for fault diagnosis [10-11].

Traditional power equipment modeling relies on engineers' experience and manual calculations, which is inefficient and prone to errors. In recent years, the application of artificial intelligence technology in the power system field has become increasingly widespread, providing new insights into addressing the aforementioned issues [12]. Deep learning technology has demonstrated strong capabilities in image recognition and pattern recognition, laying the technical foundation for automatic circuit diagram recognition [13-15]. Meanwhile, advancements in modern numerical computation methods and multi-physics simulation platforms have made automated modeling and verification of complex circuits feasible. However, existing circuit diagram recognition research has primarily focused on general-purpose circuit applications, lacking comprehensive solutions tailored to the

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specific needs of power equipment fault diagnosis [16-19]. For example, Pavithra et al.'s method for recognizing hand-drawn electronic components using the ORB algorithm primarily focuses on component recognition and digitization, suitable for single-symbol recognition [16]. Chen et al. explored the integration of power electronics and artificial intelligence, proposing the PowerVision framework for circuit diagram recognition and netlist generation, and validated circuit simulations by invoking LPSpice. However, their research focused on verifying the accuracy of circuit designs, primarily serving the needs of power electronics circuit design and analysis [17]. Charles et al. focused on the mathematical representation of circuit diagrams, achieving conversion from images to SPICE netlists, which has certain engineering application value but remains limited to basic circuit simulation verification [18].

This paper proposes an automatic modeling method for power equipment equivalent circuits based on intelligent image recognition, specifically tailored to the practical engineering needs of power equipment modeling. The main innovations of this system include: (1) Developing a deep learning recognition model specifically designed for power equipment circuit diagrams, capable of accurately identifying power components and repetitive unit structures; (2) Establishing a complete automated workflow from image recognition to frequency domain analysis, where the system can automatically generate KCL and KVL equations and convert them into state-space form, automatically solve frequency domain impedance characteristics, and provide users with complete frequency domain characteristic analysis results by simply uploading the circuit diagram; (3) Designing a netlist-to-JSON conversion mechanism, enabling the automatic generation of netlists and conversion scripts to call Simulink in the MATLAB environment for automatic simulation model construction. This module can be extended to apply to repetitive unit structures with dozens or even hundreds of levels, enabling rapid construction of simulation models and significantly improving the efficiency of modeling complex power equipment. This paper will use the equivalent circuit of a transformer winding as an example (as shown in Figure 1) to specifically illustrate the system's workflow.

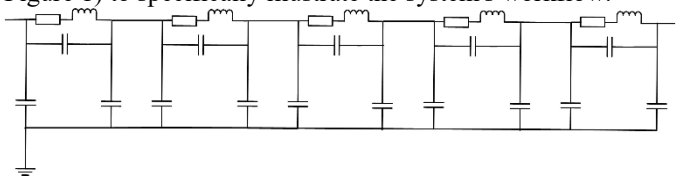


Fig. 1. Equivalent circuit of a winding transformer

II. System Architecture and Methods

This paper proposes an automatic modeling and simulation system for power equipment equivalent circuits based on intelligent image recognition. Adopting a modular design concept, the system can be decomposed into five core processing modules: circuit element recognition module, node detection and numbering module, netlist automatic generation module, equation writing and solving module, and simulation interface module. The entire system is based on intelligent design principles. Users only need to provide a circuit diagram containing repetitive units, and the system can automatically complete the entire process from image recognition to solving circuit frequency domain characteristics to constructing the simulation model.

A. Deep Learning-Based Circuit Component Recognition

Due to the limited availability of publicly accessible circuit diagram datasets and the high data requirements for training deep learning models, this module employs a hybrid data strategy to address this issue. First, we utilized circuit component image resources from the PowerVision project as a foundation to automatically generate 3,000 synthetic circuits [17]. Second, we obtained 70 real circuit diagrams from textbooks and academic papers. Finally, we adopted a three-stage progressive training strategy based on the YOLOv11 framework [19]: Stage 1 used pure synthetic data for pre-training (3,000 images, 200 epochs, lr=0.01) to quickly establish a foundational feature representation; Stage 2 employed mixed data training (real data + synthetic data, 200 epochs, lr=0.005); Stage 3 involves fine-tuning with real data (real data + a small amount of synthetic data, 200 epochs, lr=0.001). Additionally, to enhance the model's robustness under varying image quality conditions, we applied data augmentation techniques, including noise addition and rotation. Figure 2 illustrates the component recognition results.

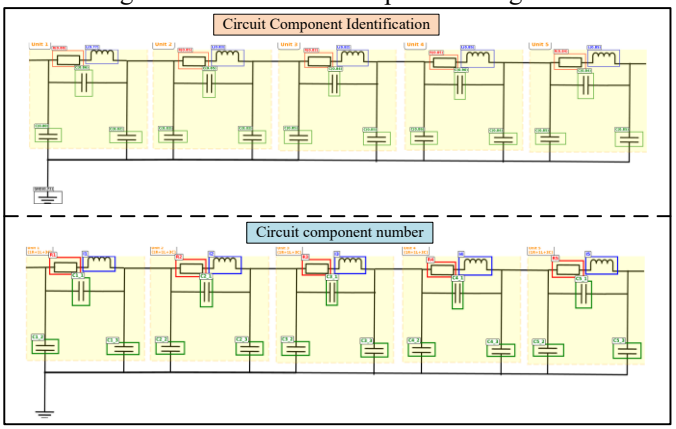


Fig. 2. Component identification results and numbers

B. Node Detection and Numbering Module

This module performs intelligent numbering based on the detection results from *A*, while identifying nodes and connection lines in the circuit. Given the repetitive unit structure characteristic of equivalent circuits in power equipment, the system adopts a unified numbering logic from left to right and top to bottom. For individual components, simplified labels are used (e.g., R_1, L_1, C_1), while multiple components of the same type are assigned hierarchical numbers (e.g., $C_{1\text{-}1}$, $C_{1\text{-}2}$, $C_{1\text{-}3}$). To detect lines and nodes in the circuit diagram, this paper employs a progressive probabilistic Hough's transform (PPHT) to identify connection lines in the circuit [20-22], with node recognition achieved through the intersection points of horizontal and vertical lines. Figure 3 illustrates the results of node detection and numbering.

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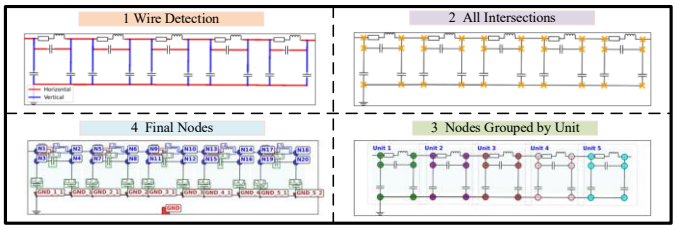


Fig. 3. Circuit diagram node detection and numbering results display

C. *Automatic Netlist Generation Module*

The generation of the netlist depends on the connection relationships between components in the circuit. Specifically, we analyze each target component individually and then use coordinate geometry to find the connection nodes. Based on the identified nodes and connection relationships, the system constructs a standard node-component association matrix. The rows of the matrix represent nodes, the columns represent components, and the matrix values 1 represent connections and 0 represent no connections. Figure 4 shows the visualization of the generated association matrix. This module achieves fully automatic conversion from image to standard netlist, providing an accurate and reliable circuit topology description for subsequent mathematical modeling and simulation analysis.

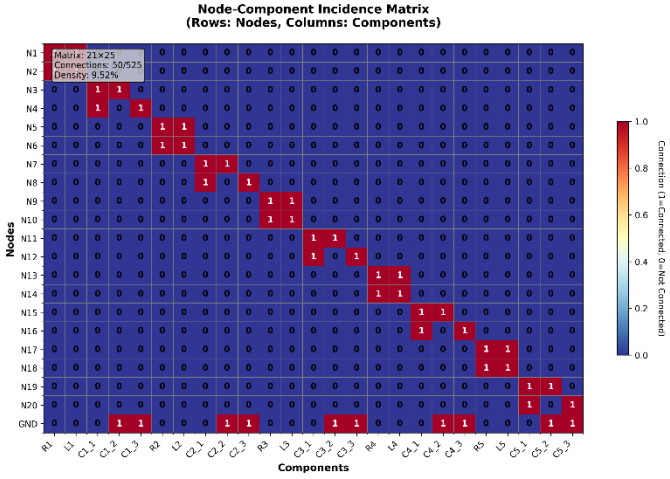


Fig. 4. Node-component incidence matrix

D. *Equation Writing and Solving Module*

This module fully utilizes the node connection relationships and component constitutive relationships obtained from the *C* module to automatically establish a complete set of KCL and KVL equations based on the circuit topology.

Automatic Generation of KCL Equations: The system defines explicit rules for current direction and symbol notation. For each node, adjacent connections are identified using position and line detection, resulting in branch currents denoted as I_Nx_Ny. The current direction is determined by node numbers: if x is less than y, I_Nx_Ny is outgoing; if x is greater than y, I_Nx_Ny) is incoming. The current I_Nx_GND is consistently defined as flowing from the main node to ground, classified as outgoing. Node N1 is assigned a default incoming current I_0. The system parses node adjacency relationships from the netlist to identify adjacent connections and generates standard KCL equations using the convention that incoming currents are positive and outgoing currents are negative.

Automatic Generation of KVL Equations: The system determines connection properties between nodes using node adjacency relationships and component connection data from module *C*, then generates the corresponding KVL equations. For node pairs connected by components, the system applies constitutive relationships according to component type. Resistors follow Ohm's law: $V_{Nx} - V_{Ny} = I_N_x_N_y \times R$. Inductors follow Faraday's law: $V_{Nx} - V_{Ny} = L \times d(I_N_x_N_y)/dt$. Capacitors establish both voltage relationships, $V_{Nx} - V_{Ny} = V_C$, and current relationships, $I_N_x_N_y = C \times d(V_C)/dt$. For node pairs connected only by wires, the system generates the equipotential constraint $V_{Nx} = V_{Ny}$. Ground connections use a unified reference potential $V_{GND} = 0$, and all voltage relationships involving ground are represented as the main node voltage equaling the corresponding component voltage drop.

This module converts topological connections into mathematical equations, establishing a comprehensive circuit equation foundation for state-space modeling and frequency-domain analysis. Figure 5 presents the KCL and KVL equations generated by this module. By integrating netlist information from module *C* with the generated equations, the standard state-space form is obtained. The impedance spectrum $Z(s) = V_{in}(s)/I_{in}(s)$ is calculated using transfer functions. The results are detailed in the experimental section.

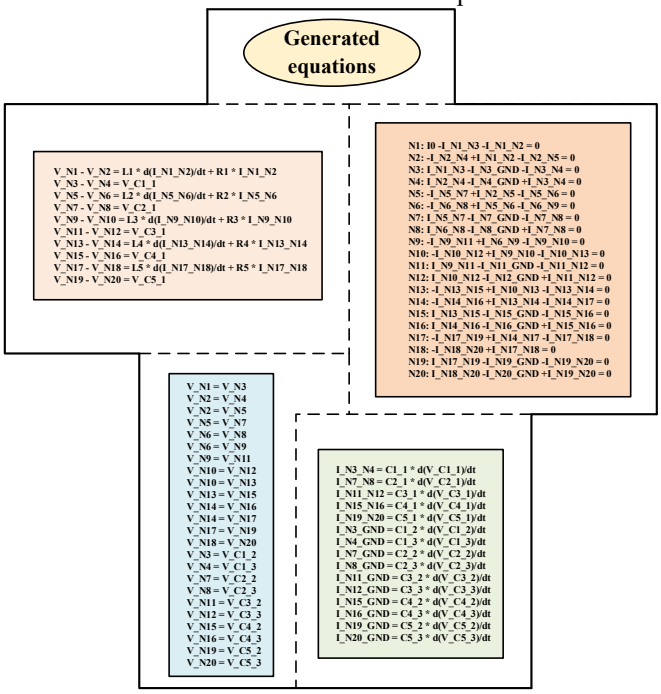


Fig. 5. Generated equations (including KCL, KVL, equipotential constraints, and capacitance-current relationships)

E. *Simulink simulation interface Module*

This module automates the conversion from topological connections to mathematical equations, providing a complete and accurate circuit equation foundation for subsequent state-space modeling and frequency-domain analysis. Figure 6 shows the KCL and KVL equations generated by this module.

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Furthermore, by combining the netlist information in *C* with the generated KCL and KVL equations, we obtain the standard state-space form. According to state-space theory, the system selects the voltage across the capacitor and the current through the series inductor as state variables. As shown in the circuit diagram in Figure 7 (ground resistance is considered in the calculations). The impedance spectrum Z(s) = Vin(s)/Iin(s) is calculated using the transfer function, and the results will be presented in the experimental section.

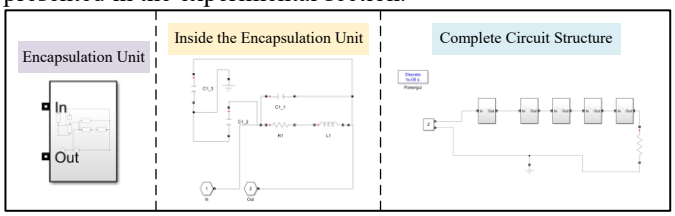


Fig. 6. Circuit diagram automatically constructed in Simulink

The *C* module stores netlist information in a JSON file format. Loading this file into MATLAB/Simulink enables the extraction of the node-component association matrix, which automatically identifies component connections for model construction. The conversion process includes a parameter input interface that permits dynamic specification of component values, such as R_s, L_s, C_g, and C_{sh}. To address the repetitive nature of circuit units, only the first basic circuit unit is extracted and encapsulated as a subsystem after reading the JSON file. Users can assemble multi-unit circuits by specifying the required number of circuit units and incorporating additional modules, including voltage sources and ground resistors. This approach is scalable to circuits containing numerous repetitive units and substantially enhances the efficiency of modeling complex circuits.

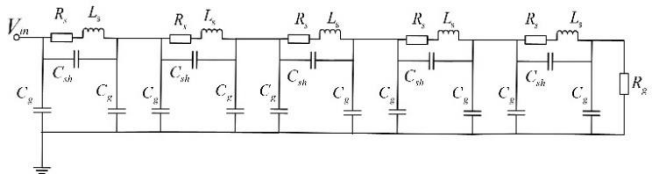


Fig. 7. Equivalent circuit considering ground resistance

III. Experimental Results and Verification

To verify the accuracy and reliability of our proposed circuit diagram intelligent recognition and modeling system, we conducted a comparative analysis of theoretical calculations, manual simulations, and automatic modeling.

1. Theoretical calculation results: Based on the state-space modeling module (Module *D*), the impedance spectrum $Z(s) = V_{in}(s)/I_{in}(s)$ is calculated using the transfer function $H(s)$.
2. Multisim manual simulation: In the Multisim environment, manually construct the equivalent circuit of the transformer winding shown in Figure 7 and perform frequency domain analysis.
3. Simulink automatic modeling: Using the *E* module proposed in this paper, automatically construct a simulation model in Simulink based on a JSON file and perform frequency domain analysis.

All three methods use the same circuit parameters and analysis conditions, as shown in Table 1.

TABLE I
SPECIFIC PARAMETER SETTINGS

Circuit parameters	R_s	L_s	C_g	C_{sh}	R_g
	1.2Ω	180e-6H	0.013e-9F	3e-9F	50Ω
Frequency range	1kHz–1GHz				

First, we present the results obtained by solving the state space equations in Module *D*, as shown in Figure 8:

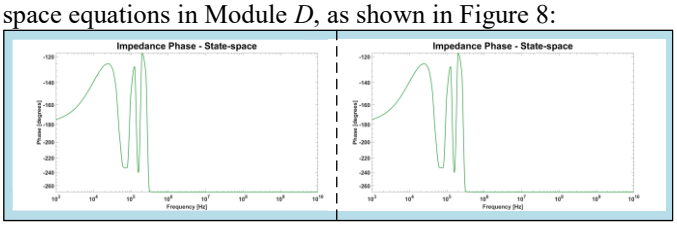


Fig. 8. Module D (state space equation) solution results

Secondly, we show the results of the Simulink simulation model automatically constructed by module E, as shown in Figure 9.

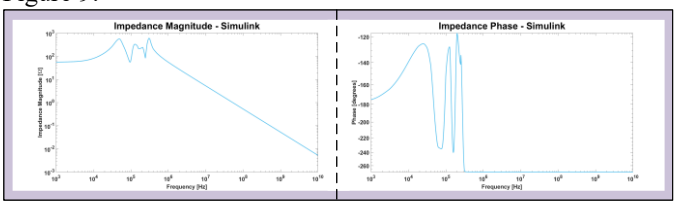


Fig. 9. Simulation results of Module E (automatically constructed Simulink model)

In order to further verify the accuracy of the system, we manually constructed the circuit diagram shown in Figure6 in Simulink, as shown in Figure 10.

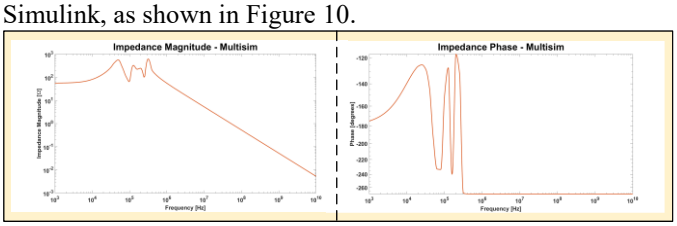


Fig. 10. Multisim simulation results

For ease of viewing, we plotted the results of solving the state-space equations alongside the simulation results from Multisim and Simulink on the same graph. As shown in Figure 11, the three curves are in near-perfect alignment, confirming that our model can be implemented simply by uploading the circuit diagram: component identification and circuit detection, generation of a netlist containing component location information, formulation of KCL and KVL equations, and automated construction of simulation models through state-space equation solutions for high-frequency impedance spectra. This system can be scaled up to handle complex circuit calculations and simulation model construction, providing convenience for power equipment modeling.

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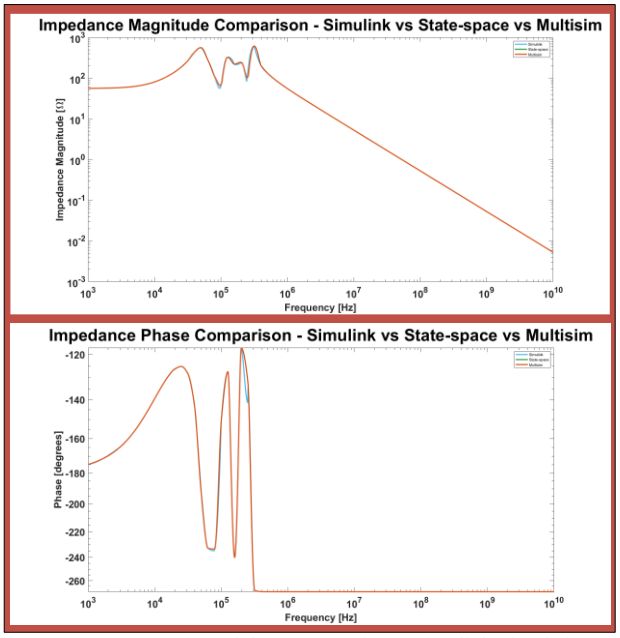


Fig. 11. Summary of spatial state equations, Simulink simulation, and Multisim simulation results

IV. CONCLUSION

This paper proposes an automatic modeling system for power equipment equivalent circuits based on intelligent image recognition, integrating deep learning technology with classical circuit theory to achieve a complete automated process from circuit images to frequency domain characteristic analysis. The system integrates five core functional modules: circuit element recognition module, node detection and numbering module, netlist automatic generation module, equation writing and solving module, and simulation interface module, the system's effectiveness was validated using the equivalent circuit of a transformer winding as an example. Experimental results show that the frequency domain impedance characteristics obtained through intelligent computation are highly consistent with the two simulation methods, fully demonstrating the system's accuracy and reliability. This successfully achieves a fully automated workflow from uploading circuit diagrams to circuit recognition and analysis, and then to simulation model construction.

The main contribution of this study lies in the successful application of artificial intelligence technology to the field of power equipment modeling, providing automated solutions for power system analysis, fault diagnosis, and equipment health monitoring. The system achieves the functionality of completing a full frequency domain analysis simply by inputting a circuit diagram through image recognition technology, significantly improving modeling efficiency, particularly demonstrating a clear technical advantage when handling large-scale repetitive unit circuits.

However, the current system has certain limitations. The system primarily focuses on modeling equivalent circuits that do not consider mutual inductance, which restricts its application scope in complex power equipment. Future work will focus on researching automated processing methods for mutual inductance modeling and developing a more comprehensive automated modeling system.

V. REFERENCES

[1] H. Henao, G. Capolino, M. Fernandez-Cabanas, F. Filippetti, C. Bruzzese, and E. Strangas, "Trends in fault diagnosis for electrical machines: A review of diagnostic techniques," *IEEE Industrial Electronics Magazine*, vol. 8, no. 2, pp. 31–42, 2014.

[2] Y. B. Shao, "Fault analysis and processing for large generator stator winding," *Large Electric Mach. Hydraulic Turbine*, vol. 2014, no. 6, pp. 27–31, Nov. 2014.

[3] F. R. Blánquez, C. A. Platero, E. Rebollo, and F. Blánquez, "Evaluation of the applicability of FRA for inter-turn fault detection in stator windings," in *Proc. 9th IEEE Int. Symp. Diagnostics Electric Mach., Power Electron. Drives*, Valencia, Spain, 2013, pp. 177–182.

[4] Y. Chen, Z. Zhao, Y. Yu, W. Wang, and C. Tang, "Understanding IFRA for detecting synchronous machine winding short circuit faults based on image classification and Smooth Grad-CAM++," *IEEE Sensors J.*, vol. 23, no. 3, pp. 2422–2432, Feb. 2023.

[5] F. Perisse, P. Werynski, and D. Roger, "A new method for AC machine turn insulation diagnostic based on high frequency resonances," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 14, no. 5, pp. 1308–1315, Oct. 2007.

[6] L. Wang, C. Ngai-Man Ho, F. Canales, and J. Jatskevich, "High-frequency modeling of the long-cable-fed induction motor drive system using TLM approach for predicting overvoltage transients," *IEEE Trans. Power Electron.*, vol. 25, no. 10, pp. 2653–2664, Oct. 2010.

[7] Y. Chen, Z. Zhao, Y. Yu, Y. Guo, and C. Tang, "Improved interpretation of impulse frequency response analysis for synchronous machine using life long learning based on iCaRL," *IEEE Trans. Instrum. Meas.*, vol. 72, Art. no. 3531510, pp. 1–10, 2023.

[8] Z. Zhao, C. Yao, C. Li, and S. Islam, "Detection of power transformer winding deformation using improved fra based on binary morphology and extreme point variation," *IEEE Trans. Ind. Electron.*, vol. 65, no. 4, pp. 3509–3519, Apr. 2018.

[9] C. A. P. Gaona, F. Blázquez, P. Frías, and M. Redondo, "A novel rotor ground-fault-detection technique for synchronous machines with static excitation," *IEEE Trans. Energy Convers.*, vol. 25, no. 4, pp. 965–973, Dec. 2010.

[10] Z. Zhao, Y. Chen, Y. Yu, M. Han, C. Tang, and C. Yao, "Equivalent broadband electrical circuit of synchronous machine winding for frequency response analysis based on gray box model," *IEEE Trans. Energy Convers.*, vol. 36, no. 4, pp. 3512–3521, Dec. 2021.

[11] C. A. Platero, F. Blázquez, P. Frías, and M. Pardo, "New on-line rotor ground fault location method for synchronous machines with static excitation," *IEEE Trans. Energy Convers.*, vol. 26, no. 2, pp. 572–580, Jun. 2011.

[12] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015.

[13] G. Linan-Cembrano, N. Lourenço, N. Horta, and J. M. de la Rosa, "Design automation of analog and mixed-signal circuits using neural networks—a tutorial brief," *IEEE Trans. Circuits Syst. II*, vol. 71, no. 3, pp. 1677–1682, 2024.

[14] J. Yang, K. Qiao, S. Shi, B. Song, J. Chen, and B. Yan, "Aem-pcb reverser: Circuit schematic generation in pcb reverse engineering using reinforcement learning based on aesthetic evaluation metric," *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.*, vol. 43, no. 5, pp. 1608–1612, 2024.

[15] H. Wang, K. Wang, J. Yang, L. Shen, N. Sun, H.-S. Lee, and S. Han, "Gcn-rl circuit designer: Transferable transistor sizing with graph neural networks and reinforcement learning," in *2020 57th ACM/IEEE Design Automation Conf. (DAC)*, 2020, pp. 1–6.

[16] C. R. Kelly and J. M. Cole, "Digitizing images of electrical-circuit schematics," *APL Mach. Learn.*, vol. 2, no. 1, pp. 016109, Mar. 2024.

[17] M. Chen and D. C. Cheng, "Power Electronics Turing Test: A Path Toward Strong AI in Power Electronics," in *2024 IEEE Workshop on Control and Modeling for Power Electronics (COMPEL)*, Lahore, Pakistan, 2024, pp. 1–8.

[18] S. Pavithra, N. K. Shreyashwini, H. S. Bhavana, G. Nikhitha, and T. Kavitha, "Hand-Drawn Electronic Component Recognition Using ORB," *Procedia Computer Science*, vol. 218, pp. 504–513, 2023.

[19] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *Proc. IEEE Conf. Computer Vision Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016, pp. 779–788.

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[20] C. Galamhos, J. Matas, and J. Kittler, "Progressive probabilistic Hough transform for line detection," in *Proc. 1999 IEEE Computer Society Conf. Computer Vision Pattern Recognition*, 1999, pp. 554–560.
[21] R. O. Duda and P. E. Hart, "Use of the Hough transformation to detect lines and curves in pictures," *Commun. ACM*, vol. 15, no. 1, pp. 11–15, 1972.
[22] P. Hough, "Method and means for recognizing complex patterns," U.S. Patent 3,069,654, Dec. 18, 1962.

VI. BIOGRAPHIES

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